A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

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1 Introduction

Irradiation-assisted stress corrosion cracking (IASCC) is a major degradation mode of austenitic stainless steel core internals in light-water reactors [25]. In irradiated stainless steels exposed to high-temperature water, intergranularoxidation weakens grain boundaries and precedes intergranular crack initiation [24], and grain-boundary oxidation has been identified as a necessary precursor of IASCC at low doses [16]. In 304 stainless steel (304SS) exposed to pressurized water reactor (PWR) primary water, the extent of this oxidation depends on the chemical changes introduced by prior irradiation. Deng et al. [5] found that increasing the proton irradiation dose produced deeper intergranular oxidation in 304 nuclear-grade stainless steel after water exposure, accompanied by a decrease in the Cr content of the intergranular oxide. Cr depletion is not the only chemical change that matters. Using model alloys with Si and Cr contents chosen to mimic the segregated boundary, Du et al. [7] showed that Si enrichment promotes intergranular oxidation and cracking even when Cr is not depleted: Si forms SiO_2 that is unstable in high-temperature water and dissolves preferentially, leaving a porous oxide that provides pathways for oxygen diffusion along grain boundaries. These observations connect irradiation to both the chemistry and the penetration depth of the intergranular oxide. Turning them into a predictive relationship requires a description of how the changing oxide composition and structure affect oxygen transport and, consequently, the advance of the oxidation front.

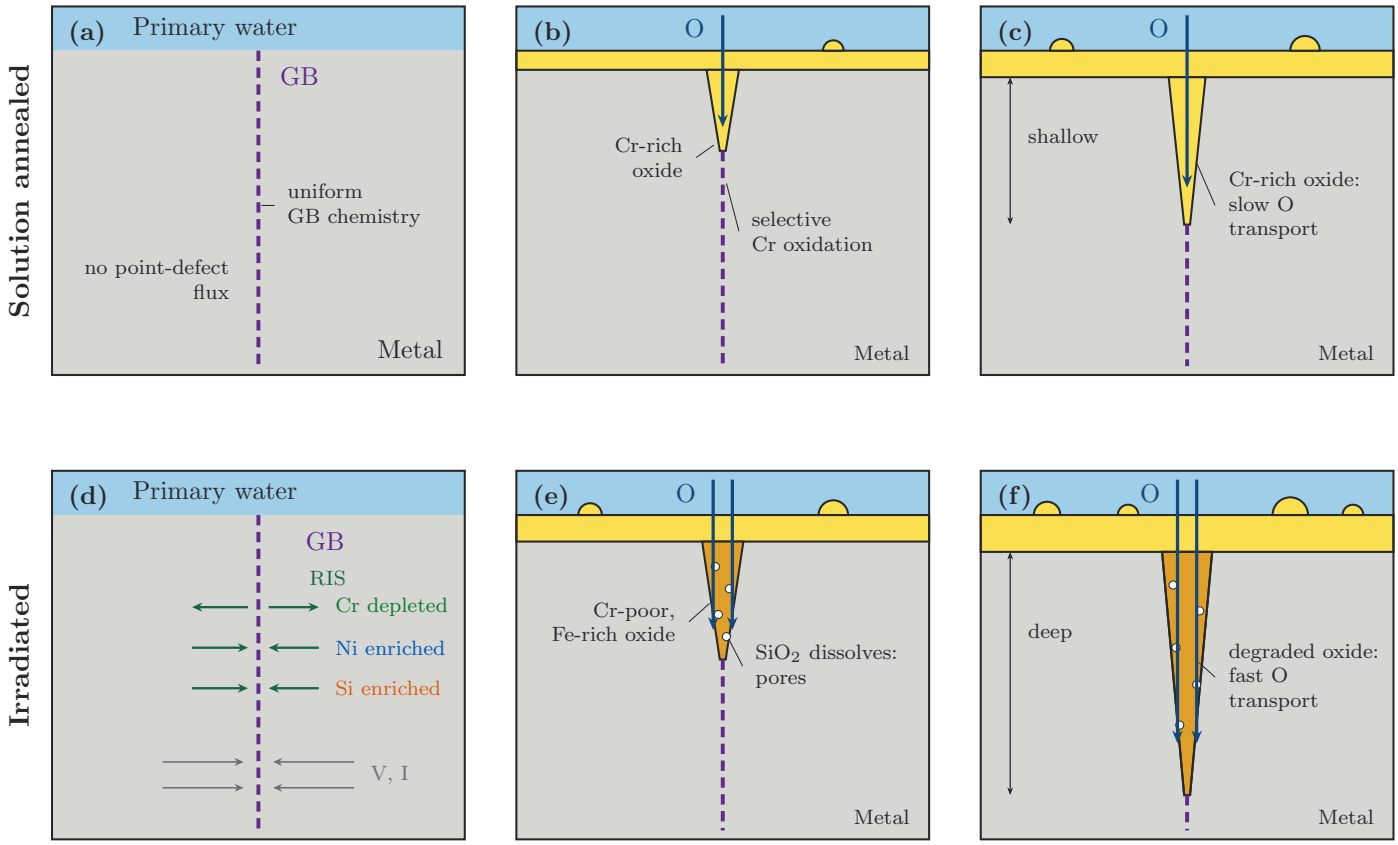


Figure 1: Physical picture of intergranular oxidation in primary water without (a–c) and with (d–f) prior irradiation. (a) Solution-annealed boundary with uniform chemistry. (b) Oxygen entering at the boundary mouth oxidizes Cr selectively and forms a Cr-rich intergranular oxide. (c) The Cr-rich oxide limits oxygen transport along the boundary and the penetration stays shallow. (d) Irradiation drives point defects to the boundary and the resulting radiation-induced segregation depletes Cr and enriches Ni and Si. (e) On exposure the Cr-depleted boundary forms a Cr-poor, Fe-rich oxide, and the segregated Si oxidizes and dissolves, leaving pores. (f) The degraded oxide transports oxygen faster along the boundary and the oxide penetrates deeper.

Radiation-induced segregation (RIS) establishes the local alloy composition from which the intergranular oxide develops. Coupling between solute and irradiation-generated point-defect fluxes redistributes alloying elements near grain boundaries. Measurements on irradiated 304 nuclear-grade stainless steel reveal Cr depletion and Ni and Si enrichment at the grain boundary [3]. Subsequent oxidation selectively consumes alloying elements, changing both the oxide composition and the adjacent metal, where Ni enrichment has also been observed [3]. The lower Cr content of the oxide formed on irradiated boundaries has been associated with reduced resistance to oxygen diffusion [5].

Si oxidation and dissolution alter this transport resistance by changing the structure through which oxygen moves. The role of Si was established by Du et al. [7] with model alloys of systematically varied Cr and Si contents in hydrogenated water, which allowed the contributions of Cr depletion and Si enrichment to be separated. They attributed the greater oxide degradation and cracking susceptibility at higher Si content to dissolution of SiO_2 in the inner oxide and the resulting porous, defective film, and their STEM analysis of intergranular oxides indicated faster grain-boundary diffusion at higher Si content. Related microscopy on proton-irradiated 316L identified Si enrichment at oxide tips and porosity behind them, consistent with preferential Si oxidation followed by dissolution [16]. Although that alloy differs from 304SS in bulk composition, the sequence itself is not alloy-specific: oxidation and dissolution of segregated Si appears as a step in the IASCC mechanism summarized by Was et al. [24] for austenitic stainless steels generally. Together, these studies provide a basis for including dissolution-generated free volume in the transport description. With the dependence on Cr content, they motivate representing the evolving oxide composition and free volume through a single effective oxygen diffusivity, which connects the segregation of multiple elements to a common quantity governing oxygen transport.

Each half of this sequence has been modeled separately, and the two halves are not equally developed. The segregation half is settled enough that the grain boundary composition can be computed rather than only measured. Since the inverse-Kirkendall and rate-theory descriptions of RIS in concentrated alloys [17, 26], the composition profile at a grain boundary has been obtained from the coupled fluxes of point defects and alloying elements, with Fe–Cr–Ni parameterizations that reproduce the measured dose and temperature dependence of Cr depletion and Ni enrichment [1]. Current models go further and resolve the sink itself, predicting segregation at grain boundaries

and dislocation cell walls [14] and the conditions under which Cr switches from enrichment to depletion [19]. All of these models return the segregation profile as their final output; none includes how the modified boundary composition affects the subsequent grain-boundary oxidation.

On the oxidation side, the dependence of intergranular penetration depth on dose and exposure time has been characterized experimentally [4, 5], but existing kinetic models of grain-boundary oxidation do not respond to the grain boundary chemistry. Models that resolve oxygen transport along the grain boundary treat the oxygen diffusivity as a single constant: in analytical or semi-analytical solutions that give the penetration depth in closed form [8, 10, 27], in finite-element descriptions in which it is fitted to measured oxide depths through a parabolic law [15, 28] or prescribed together with a constant reaction rate [2], and in phase-field simulations of a single stoichiometric chromia intrusion [6, 21]. In none of these does the diffusivity depend on the composition of the oxide that has formed. Where the oxide identity does control transport, in multilayer descriptions of planar scales [12, 13] and in the mechanistic treatment of uniform corrosion in high-temperature water [22], the controlling phase is fixed at the outset rather than selected by the local alloy composition. The boundary composition enters only after conversion into an empirical parameter, as a Cr-content-dependent dissolution rate in sensitized steels [20] or as a sensitization index that increases with neutron fluence in the slip-oxidation prediction of crack growth [11]; crack-growth models that require a grain-boundary oxidation law borrow rate constants measured on bulk surfaces [23]. Thermodynamic selection of the oxide phases from the local alloy composition has been implemented for a Ni-base superalloy [9], but with a constant oxygen diffusivity, the bulk composition at the boundary, and without the Cr oxide that limits transport. No model takes the irradiation-modified boundary composition as input and lets the resulting oxide composition set the oxygen transport that governs further penetration.

To our knowledge, no model connects the segregated composition to the advance of the oxidation front through a transport quantity computed from both. Establishing this connection requires tracking how the segregated alloy supplies the interfacial reactions, how those reactions change the oxide composition, and how the resulting diffusivity affects subsequent oxygen delivery. Segregation profiles, oxide composition, interfacial Ni enrichment, and penetration depth then provide complementary observables for assessing the coupling.

Here, we develop a coupled RIS and grain-boundary oxidation model for 304SS, represented by the Fe–Cr–Ni–Si system. The model links defect-mediated solute redistribution and selective interfacial consumption to an effective oxygen diffusivity, D_{eff}, determined by the evolving oxide composition and free volume. Within this description, increasing the Cr content of the oxide lowers the effective diffusivity, whereas Si oxidation followed by complete dissolution contributes free volume that enhances transport. The same D_{eff} enters both oxygen transport along the grain boundary and oxygen delivery across the oxide to the metal interface. The model therefore connects RIS to oxidation kinetics through the resulting change in oxygen transport, with metal consumption feeding back on the local composition. The complete-dissolution treatment is a simplification motivated by the mechanism discussed by Du and co-workers [7]. We examine the dose and exposure-time dependence of oxide penetration under sequential irradiation and water exposure, using separate calibration of the RIS and oxidation components and a common set of oxidation parameters across doses. The framework also permits predictions of concurrent RIS and oxidation within the same transport description.

2 Model Formulation and Calibration

The model consists of two components: a radiation-induced segregation (RIS) model for the evolving GB chemistry and a reduced-dimensional oxidation model for oxygen transport, consumption, and oxide growth. The RIS-predicted Cr, Ni, Si, and Fe concentrations provide the local alloy composition for the oxidation model. Oxygen transport along the boundary is solved for the cross-section-averaged concentration of the oxidized zone, and the delivery of oxygen to the oxide/metal interface is closed by the fundamental mode of the steady diffusion equation in the cross-section. The oxide composition determines the effective oxygen diffusivity through an effective-medium treatment, which controls the local oxygen consumption and oxide growth rate. The formulations of the two components are described below.

Figure 2 summarises the model. The computational domain, Fig. 2a, is a grain boundary that opens onto the surface exposed to the primary water. Oxygen enters at the mouth through the surface film and migrates along the boundary within the oxidized zone, whose half-width grows as the oxide thickens. Only the half-domain on one side of the boundary is simulated, the

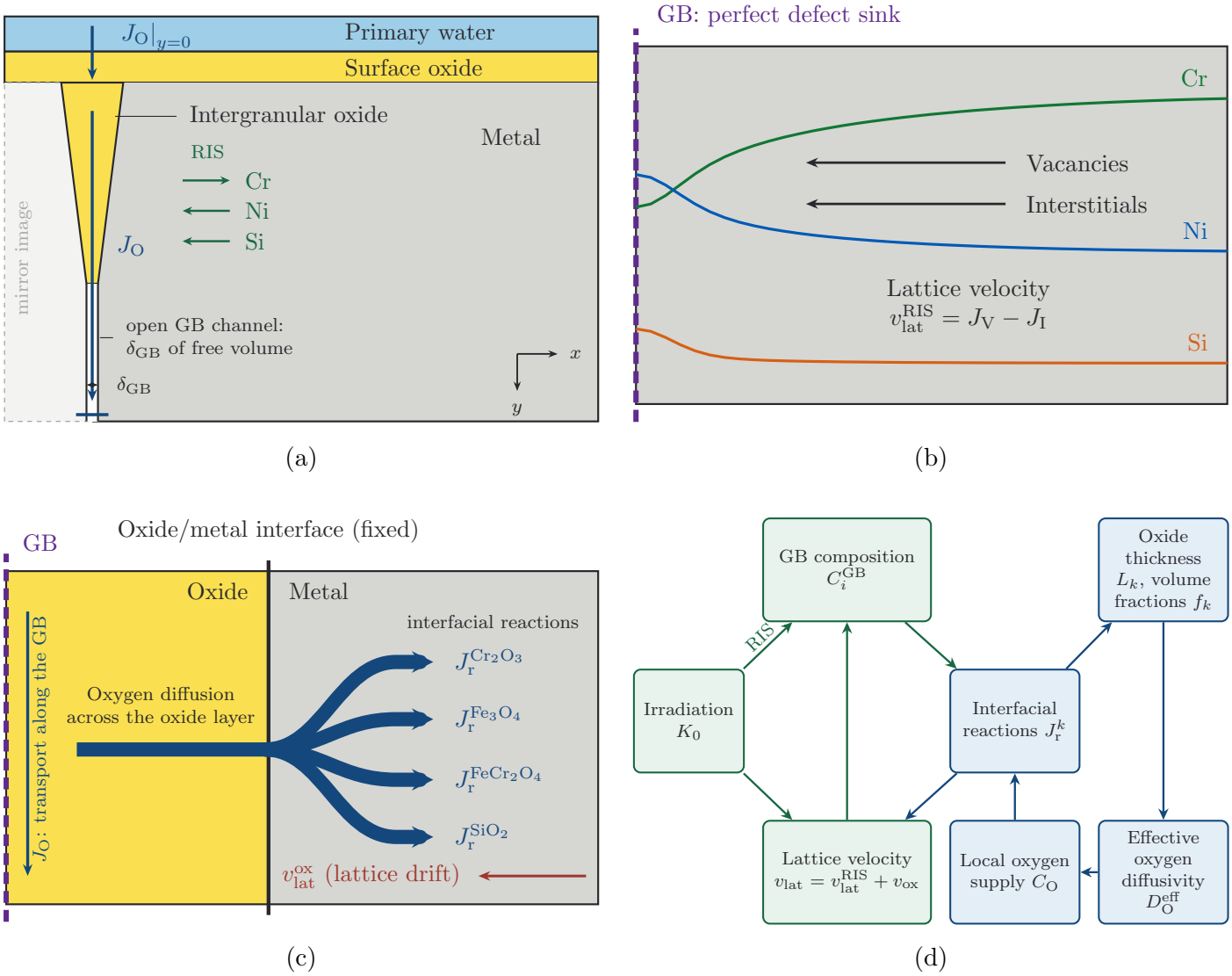


Figure 2: Overview of the model: (a) physical picture and computational domain, (b) radiation-induced segregation, (c) oxygen transport across the oxide and the interfacial reactions, and (d) coupling between the two components.

other half following by symmetry.

The boundary acts as a sink of finite efficiency for the vacancies and self-interstitials produced by irradiation, Fig. 2b. The resulting redistribution depletes Cr and enriches Ni and Si at the boundary, and the imbalance between the two defect fluxes gives the segregation contribution $\mathbf{v}_{\mathrm{lat}}^{\mathrm{RIS}} = \mathbf{J}_{\mathrm{V}} - \mathbf{J}_{\mathrm{I}}$ to the lattice velocity; the profiles shown are calculated, mirrored about the boundary and convolved with a 1 nm FWHM Gaussian. At a given depth, Fig. 2c, the oxygen delivered along the boundary reaches the advancing oxide/metal interface through the oxide cross-section and is partitioned there among the four oxide phases; the metal swept by the front is either oxidized, rejected into the metal, or carried away through the oxide, and the net removal sets the oxidation contribution v_{ox}. The two components are coupled as shown in Fig. 2d: irradiation fixes the boundary chemistry and part of the lattice velocity, the boundary chemistry controls the interfacial reaction rates, and the oxide that forms feeds back on the oxygen supply through its thickness and its effective diffusivity.

Notation. Alloying-element indices are $i, j \in \{\mathrm{Cr}, \mathrm{Ni}, \mathrm{Si}, \mathrm{Fe}\}$; oxide-phase indices are $k \in \{\mathrm{Cr_2O_3}, \mathrm{Fe_3O_4}, \mathrm{FeCr_2O_4}, \mathrm{SiO_2}\}$. Superscripts V and I denote quantities associated with vacancies and self-interstitials; the superscript GB denotes a value taken on the grain boundary. C is a site fraction, $\mathbf{J}$ a flux in the two-dimensional metal domain, and J a flux on the one-dimensional grain-boundary line. The coordinate x is normal to the boundary and y runs along it.

2.1 Radiation-Induced Segregation Model

The RIS model is based on solving a set of coupled mass-conservation equations for the alloying elements (Cr, Ni, Si, and Fe) and point defects (vacancies and self-interstitials). For each alloying element i, the concentration evolves according to

$$\frac{\partial C_i}{\partial t} = -\nabla \cdot \mathbf{J}_i. \tag{1}$$

The site fractions of vacancies and self-interstitials are governed separately by

$$\frac{\partial C_{\mathrm{V}}}{\partial t} = -\nabla\cdot\mathbf{J}_{\mathrm{V}} + \epsilon K_0 - R\,C_{\mathrm{I}}C_{\mathrm{V}}, \tag{2}$$

$$\frac{\partial C_{\mathrm{I}}}{\partial t} = -\nabla\cdot\mathbf{J}_{\mathrm{I}} + \epsilon K_0 - R\,C_{\mathrm{I}}C_{\mathrm{V}}, \tag{3}$$

where C_{V} and C_{I} are the vacancy and self-interstitial site fractions, ϵ is the damage efficiency, K_0 is the radiation dose rate, and R is the vacancy–interstitial recombination rate coefficient. The flux of each alloying element carried by each defect mechanism is

$$\mathbf{J}_i^{\mathrm{V}} = -\sum_j L_{ij}^{\mathrm{V}}\,\nabla(\mu_j - \mu_{\mathrm{V}})\,, \tag{4}$$

$$\mathbf{J}_i^{\mathrm{I}} = -\sum_j L_{ij}^{\mathrm{I}}\,\nabla(\mu_j + \mu_{\mathrm{I}})\,, \tag{5}$$

with the Onsager coefficients

$$L_{ii}^{\mathrm{V}} = \frac{C_i D_i^{\mathrm{V}} C_{\mathrm{V}}}{k_B T}\left(1 + \frac{1-f_0^{\mathrm{V}}}{f_0^{\mathrm{V}}}\,\frac{C_i D_i^{\mathrm{V}}}{\sum_m C_m D_m^{\mathrm{V}}}\right), \tag{6}$$

$$L_{ij}^{\mathrm{V}} = L_{ji}^{\mathrm{V}} = \frac{1-f_0^{\mathrm{V}}}{f_0^{\mathrm{V}}}\,\frac{C_i D_i^{\mathrm{V}}\,C_j D_j^{\mathrm{V}}\,C_{\mathrm{V}}}{k_B T\,\sum_m C_m D_m^{\mathrm{V}}}, \qquad i \neq j, \tag{7}$$

$$L_{ii}^{\mathrm{I}} = \frac{C_i D_i^{\mathrm{I}} C_{\mathrm{I}}}{k_B T}\left(1 + \frac{1-f_0^{\mathrm{I}}}{f_0^{\mathrm{I}}}\,\frac{C_i D_i^{\mathrm{I}}}{\sum_m C_m D_m^{\mathrm{I}}}\right), \tag{8}$$

$$L_{ij}^{\mathrm{I}} = L_{ji}^{\mathrm{I}} = \frac{1-f_0^{\mathrm{I}}}{f_0^{\mathrm{I}}}\,\frac{C_i D_i^{\mathrm{I}}\,C_j D_j^{\mathrm{I}}\,C_{\mathrm{I}}}{k_B T\,\sum_m C_m D_m^{\mathrm{I}}}, \qquad i \neq j, \tag{9}$$

where D_i^{V} and D_i^{I} are the mechanism-specific diffusion coefficients of element i, and f_0^{V} and f_0^{I} are the corresponding correlation factors. The net defect fluxes follow from conservation of lattice sites,

$$\mathbf{J}_{\mathrm{V}} = -\sum_i \mathbf{J}_i^{\mathrm{V}}, \tag{10}$$

$$\mathbf{J}_{\mathrm{I}} = \sum_i \mathbf{J}_i^{\mathrm{I}}, \tag{11}$$

and give rise to a lattice velocity

$$\mathbf{v}_{\mathrm{lat}}^{\mathrm{RIS}} = \mathbf{J}_{\mathrm{V}} - \mathbf{J}_{\mathrm{I}}. \tag{12}$$

Grain-boundary sink condition. The boundary is treated as a sink of finite efficiency for both point defects. At the boundary plane the defect fluxes obey the Robin conditions

$$\mathbf{J}_{\mathrm{V}}\cdot\hat{\mathbf{e}}_x\big|_{x=0} = k_{\mathrm{V}}^{\mathrm{GB}}\left(C_{\mathrm{V}}\big|_{x=0} - C_{\mathrm{V}}^{\mathrm{eq}}\right), \tag{13}$$

$$\mathbf{J}_{\mathrm{I}}\cdot\hat{\mathbf{e}}_x\big|_{x=0} = k_{\mathrm{I}}^{\mathrm{GB}}\left(C_{\mathrm{I}}\big|_{x=0} - C_{\mathrm{I}}^{\mathrm{eq}}\right), \tag{14}$$

where $C_{\mathrm{V}}^{\mathrm{eq}}$ and $C_{\mathrm{I}}^{\mathrm{eq}}$ are the thermal-equilibrium site fractions and $k_{\mathrm{V}}^{\mathrm{GB}}$, $k_{\mathrm{I}}^{\mathrm{GB}}$ are the boundary transfer coefficients of the two defects. The perfect-sink (Dirichlet) condition $C_{\mathrm{V}} = C_{\mathrm{V}}^{\mathrm{eq}}$, $C_{\mathrm{I}} = C_{\mathrm{I}}^{\mathrm{eq}}$ is recovered for $k^{\mathrm{GB}} \to \infty$. With the Dirichlet condition the defect gradients at the boundary, and with them both the magnitude and the width of the segregation profiles, do not converge under mesh refinement, a behaviour also reported for other Onsager-type RIS models [18]; the finite-efficiency condition removes this dependence (Supplementary Information, Sec. S1). The transfer coefficients are calibrated together with the defect-mediated transport coefficients, Sec. 2.4.1.

2.2 Grain-Boundary Oxidation Model

The oxidized zone is treated as a channel of half-width $L(y,t)$, the distance from the boundary mid-plane to the oxide/metal interface, which the oxide phases of thickness L_k fill, $L = \sum_k L_k$. The oxide is mixed locally rather than layered, so a single effective diffusivity applies across and along the boundary,

$$D_{\mathrm{O}}^{\mathrm{eff}} = \sum_k f_k\, D_{\mathrm{O}}^{k}, \tag{15}$$

with f_k the volume fraction and D_{O}^{k} the oxygen diffusivity of phase k. Integrating the two-dimensional oxygen balance over the cross-section gives, for the section-averaged concentration $\bar{C}_{\mathrm{O}}(y,t)$,

$$\frac{\partial}{\partial t}\big(L\,\bar{C}_{\mathrm{O}}\big) = -\frac{\partial}{\partial y}\big(L\,J_{\mathrm{O}}\big) - J_{\mathrm{r}}, \tag{16}$$

$$J_{\mathrm{O}} = -D_{\mathrm{O}}^{\mathrm{eff}}\,\frac{\partial \bar{C}_{\mathrm{O}}}{\partial y}, \tag{17}$$

where J_{r} is the oxygen flux reaching the interface. Equation (16) is exact up to terms of order $\partial L/\partial y$ and makes no assumption on the lateral concentration profile; L inside the time derivative accounts for the dilution of dissolved oxygen as the section widens (Supplementary Information, Sec. S3).

The relation between J_r and $\bar{C}_\mathrm{O}$ is obtained from the fundamental separable mode of the steady diffusion equation in the cross-section, $C_\mathrm{O} \propto \cos(\mu x/L)$, which satisfies the symmetry condition at the mid-plane and the reaction condition at the interface. It gives

$$J_\mathrm{r} = \frac{D_\mathrm{O}^\mathrm{eff}\,\mu^2\,\bar{C}_\mathrm{O}}{L}, \qquad C_\mathrm{O}^\mathrm{int} = \bar{C}_\mathrm{O}\,\mu\cot\mu, \tag{18}$$

where $C_\mathrm{O}^\mathrm{int}$ is the oxygen concentration at the interface and the dimensionless eigenvalue $\mu \in (0, \pi/2)$ is fixed at every position by the interfacial balance between oxygen supply and consumption,

$$\frac{D_\mathrm{O}^\mathrm{eff}\,\mu^2\,\bar{C}_\mathrm{O}}{L} = \sum_k J_\mathrm{r}^k\left(\bar{C}_\mathrm{O}\,\mu\cot\mu\right). \tag{19}$$

The interfacial reaction rates of the four oxide phases are

$$J_\mathrm{r}^{\mathrm{Cr_2O_3}} = k_\mathrm{r}^{\mathrm{Cr_2O_3}}\,C_\mathrm{O}^\mathrm{int}\,C_\mathrm{Cr}^\mathrm{GB}, \tag{20}$$

$$J_\mathrm{r}^{\mathrm{Fe_3O_4}} = k_\mathrm{r}^{\mathrm{Fe_3O_4}}\left(C_\mathrm{O}^\mathrm{int} - C_\mathrm{O}^\mathrm{th}\right)C_\mathrm{Fe}^\mathrm{GB}, \tag{21}$$

$$J_\mathrm{r}^{\mathrm{FeCr_2O_4}} = k_\mathrm{r}^{\mathrm{FeCr_2O_4}}\,C_\mathrm{O}^\mathrm{int}\left(\frac{1}{C_\mathrm{Fe}^\mathrm{GB}} + \frac{2}{C_\mathrm{Cr}^\mathrm{GB}}\right)^{-1}, \tag{22}$$

$$J_\mathrm{r}^{\mathrm{SiO_2}} = k_\mathrm{r}^{\mathrm{SiO_2}}\,C_\mathrm{O}^\mathrm{int}\,C_\mathrm{Si}^\mathrm{GB}, \tag{23}$$

where k_r^k is the interfacial rate constant of phase k and C_O^th is the threshold oxygen concentration for magnetite formation; their sum is the total interfacial consumption,

$$J_\mathrm{r} = J_\mathrm{r}^{\mathrm{Cr_2O_3}} + J_\mathrm{r}^{\mathrm{Fe_3O_4}} + J_\mathrm{r}^{\mathrm{FeCr_2O_4}} + J_\mathrm{r}^{\mathrm{SiO_2}}. \tag{24}$$

The left-hand side of Eq. (19) increases and the right-hand side decreases monotonically with μ, so the root is unique; it is found by Newton's method. In the reaction-limited limit, $\mu^2 \simeq kL/D_\mathrm{O}^\mathrm{eff}$ with $k = \mathrm{d}J_\mathrm{r}/\mathrm{d}C_\mathrm{O}^\mathrm{int}$, the interface sees the average concentration, $C_\mathrm{O}^\mathrm{int} \simeq \bar{C}_\mathrm{O}$; in the diffusion-limited limit $\mu \to \pi/2$, $C_\mathrm{O}^\mathrm{int} \to 0$ and $J_\mathrm{r} \to \pi^2 D_\mathrm{O}^\mathrm{eff}\bar{C}_\mathrm{O}/4L$. The closure thus spans both regimes without a prescribed transport resistance; its derivation, the neglected higher modes and its range of validity are given in the Supplementary Information, Sec. S3. The corresponding rates of metal removal follow from the stoichiometry of each phase,

$$J_\mathrm{r}^\mathrm{Cr} = \tfrac{2}{3}J_\mathrm{r}^{\mathrm{Cr_2O_3}} + \tfrac{2}{4}J_\mathrm{r}^{\mathrm{FeCr_2O_4}}, \tag{25}$$

$$J_\mathrm{r}^\mathrm{Fe} = \tfrac{3}{4}J_\mathrm{r}^{\mathrm{Fe_3O_4}} + \tfrac{1}{4}J_\mathrm{r}^{\mathrm{FeCr_2O_4}}, \tag{26}$$

$$J_\mathrm{r}^\mathrm{Si} = \tfrac{1}{2}J_\mathrm{r}^{\mathrm{SiO_2}}. \tag{27}$$

Interfacial mass redistribution. The element with the largest ratio of consumption to availability sets the rate at which the oxidation front advances into the metal,

$$v_{\mathrm{lat}}^{\mathrm{ox,max}} = \max_i \frac{J_{\mathrm{r}}^i}{C_i^{\mathrm{GB}}}, \tag{28}$$

with $J_{\mathrm{r}}^{\mathrm{Ni}} = 0$ since Ni is not oxidized. The front sweeps element i at the rate $v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}}$, of which only J_{r}^i is oxidized; the excess

$$E_i = \max\bigl(v_{\mathrm{lat}}^{\mathrm{ox,max}} C_i^{\mathrm{GB}} - J_{\mathrm{r}}^i,\, 0\bigr) \tag{29}$$

is either rejected into the metal ahead of the front or removed through the oxide, where it is retained as unoxidized solute or transported along the oxide grain boundaries to the surface [4]. The fraction removed through the oxide is

$$\lambda_i = \frac{v_{\mathrm{lat}}^{\mathrm{ox,max}}}{v_{\mathrm{lat}}^{\mathrm{ox,max}} + v_i^{\mathrm{r}}}, \tag{30}$$

where v_i^{r} is the redistribution velocity of element i, the characteristic velocity at which it can be rejected into the metal. Complete rejection ($\lambda_i = 0$) is recovered for $v_i^{\mathrm{r}} \to \infty$; for $v_i^{\mathrm{r}} \ll v_{\mathrm{lat}}^{\mathrm{ox,max}}$ the front outruns the rejection and the excess is carried into the oxide ($\lambda_i \to 1$). The net removal of element i from the metal is then

$$\tilde{J}_{\mathrm{r}}^i = J_{\mathrm{r}}^i + \lambda_i E_i, \tag{31}$$

which replaces J_{r}^i in the interfacial conditions of Sec. 2.3.1. The motivation for this treatment and its limiting behaviour are given in the Supplementary Information, Sec. S2.

2.3 Coupling Strategy and Simulation Protocol

2.3.1 Lattice Velocity and Metal Consumption

Metal atoms leave the metal domain only at the grain-boundary plane, either by oxidation or by removal through the oxide, Eq. (31). Because the metal lattice is incompressible, this removal creates no source in the interior: the entire lattice translates towards the interface at the velocity set by the net interfacial removal. Denoting by

$$v_{\mathrm{ox}}(y,t) = \sum_i \tilde{J}_{\mathrm{r}}^i = \sum_i \bigl(J_{\mathrm{r}}^i + \lambda_i E_i\bigr) \tag{32}$$

the recession rate of the metal/oxide interface, which lies between the oxidation rate $\sum_i J_{\mathrm{r}}^i$ (complete rejection) and the sweep velocity $v_{\mathrm{lat}}^{\mathrm{ox,max}}$ (complete removal), the total lattice velocity in the metal domain is

$$\mathbf{v}_{\mathrm{lat}}(x,y,t) = \mathbf{v}_{\mathrm{lat}}^{\mathrm{RIS}}(x,y,t) + v_{\mathrm{ox}}(y,t)\,\hat{\mathbf{e}}_x. \tag{33}$$

The two contributions differ in character. The segregation term is local: it is determined pointwise by the defect fluxes at (x, y). The oxidation term is not; it is evaluated at the boundary and is independent of x, so a single value applies across the whole domain at a given y. Segregation therefore redistributes the lattice internally, whereas oxidation translates it rigidly, so that the total flux of alloying element i entering Eq. (1) is

$$\mathbf{J}_i = \mathbf{J}_i^{\mathrm{V}} + \mathbf{J}_i^{\mathrm{I}} + C_i \mathbf{v}_{\mathrm{lat}}, \tag{34}$$

subject to the interfacial condition

$$\mathbf{J}_i \cdot \hat{\mathbf{e}}_x\big|_{x=0} = \tilde{J}_{\mathrm{r}}^i, \tag{35}$$

which removes from the boundary plane the atoms incorporated into the growing oxide or carried away through it.

Equations (34) and (35) collect the three ways in which an alloying element can move. The first two terms are the diffusive fluxes carried by the two defect mechanisms, Eqs. (4) and (5); the third is the drift of the entire lattice relative to the laboratory frame; the interfacial consumption enters not as a bulk term but as an outgoing flux on the boundary plane.

The two contributions to $\mathbf{v}_{\mathrm{lat}}$ have different origins but act identically on every species. $\mathbf{v}_{\mathrm{lat}}^{\mathrm{RIS}}$ arises because the vacancy and interstitial fluxes are unequal, so lattice sites are created and destroyed at different rates; v_{ox} arises because oxidation consumes metal atoms from the interfacial layer, so the lattice is displaced towards the advancing front. Because the drift term $C_i \mathbf{v}_{\mathrm{lat}}$ is proportional to the local concentration and carries no species selectivity, it does not by itself alter the relative composition; selectivity enters only through the mechanism-specific diffusivities in Eqs. (6)–(9), through the stoichiometric sinks in Eqs. (25)–(27), and through the element-specific redistribution fractions λ_i of Eq. (30).

2.3.2 Sequential Irradiation–Oxidation Protocol

The reference experiments irradiate the specimen first and expose it to the primary water environment afterwards. The model reproduces this protocol

as two consecutive stages that share the same governing equations and differ only in two switches: the dose rate K_0 and the oxygen boundary condition at the grain-boundary mouth.

In the irradiation stage the specimen is irradiated to the target dose with no oxygen present, $K_0 > 0$ and $C_{\mathrm{O}} \equiv 0$, so that Eqs. (1)–(12) govern the evolution alone and all interfacial reaction rates vanish identically. The stage is integrated to $t_{\mathrm{irr}} = \Phi/K_0$, where Φ is the nominal dose in dpa, and yields the segregated composition profiles $C_i(x, y, t_{\mathrm{irr}})$.

The oxidation stage continues from the final state of the irradiation stage: all concentration fields, including the point-defect populations, are inherited unchanged,

$$(C_i,\, C_{\mathrm{V}},\, C_{\mathrm{I}})\big|_{t=0^+} = (C_i,\, C_{\mathrm{V}},\, C_{\mathrm{I}})\big|_{t=t_{\mathrm{irr}}}\,, \qquad L_k\big|_{t=0^+} = 0, \tag{36}$$

and the exposure clock is restarted from zero. The dose rate is then set to zero, $K_0 = 0$, so that the inherited defect supersaturation decays by recombination according to Eqs. (2) and (3) without further production.

Simultaneously, the oxygen boundary condition at the mouth of the boundary is switched on. Oxygen ingress is limited by the surface film that grows on the free surface, whose resistance increases as the film thickens; this is imposed as a Robin condition with a time-decaying transfer coefficient,

$$J_{\mathrm{O}}\big|_{y=0} = h(t)\left(C_{\mathrm{O}}^{\mathrm{w}} - \bar{C}_{\mathrm{O}}\big|_{y=0}\right), \qquad h(t) = \frac{k_{\mathrm{s}}}{\sqrt{t} + \sqrt{t_0}}, \tag{37}$$

where $C_{\mathrm{O}}^{\mathrm{w}}$ is the oxygen concentration of the bulk water, k_{s} is the surface transfer coefficient, and t_0 regularizes the initial instant. The $t^{-1/2}$ decay reproduces the parabolic thickening of the surface film, so that the cumulative oxygen admitted through the mouth grows as $t^{1/2}$. The deep end of the boundary is closed, $J_{\mathrm{O}}\big|_{y=L_{\mathrm{GB}}} = 0$.

Because no point defects are produced once $K_0 = 0$, the inherited supersaturation decays on the recombination timescale, which is short compared with the exposure time; segregation therefore effectively ceases early in the exposure stage, and the composition field that the oxidation model sees is the one established during irradiation. The sequential protocol thus isolates the compositional pathway: any dose dependence of the predicted oxidation depth must be carried by the radiation-induced segregation profile inherited through Eq. (36), since neither radiolysis nor sustained irradiation-assisted transport is present.

2.3.3 Fully Coupled Irradiation–Oxidation Framework

The sequential protocol of Sec. 2.3.2 is a special case of the governing equations rather than a separate model: it is obtained by switching K_0 and the mouth boundary condition on and off in turn. Leaving both switches on throughout, $K_0 > 0$ together with the Robin condition Eq. (37), gives the fully coupled problem in which segregation and oxidation proceed simultaneously, as they do in reactor service. No additional equations or parameters are required; Eqs. (1)–(12) are integrated as a single stage.

In this mode the two processes feed back on each other. Segregation continues to modify the boundary composition while the oxidation front advances, so the composition that the interfacial reactions see evolves in time rather than being fixed at its end-of-irradiation value; conversely, the metal consumed at the front and the associated lattice drift, Eq. (32), alter the concentration field on which the segregation fluxes act. Predictions obtained in this mode are reported in Sec. 3.5; because no experiment in the present dataset imposes concurrent irradiation and exposure, this mode is used only for prediction and takes no part in the calibration of Sec. 2.4.

2.4 Model Parameterization and Calibration

The two components are parameterized and calibrated separately and against separate data. The RIS model is calibrated first; the oxidation model is then calibrated with the RIS model related parameters fixed.

2.4.1 RIS Parameterization

The RIS model requires the mechanism-specific diffusion coefficients D_i^{V} and D_i^{I} of the four alloying elements, the correlation factors f_0^{V} and f_0^{I}, the recombination coefficient R, the damage efficiency ϵ, the dose rate K_0, and the grain-boundary transfer coefficients $k_{\mathrm{V}}^{\mathrm{GB}}$ and $k_{\mathrm{I}}^{\mathrm{GB}}$ of Eqs. (13) and (14). The dose rate and irradiation temperature are those of the reference experiment. The correlation factors are taken from their crystallographic values for the fcc lattice. The damage efficiency accounts for the fraction of displacements that survive in-cascade recombination and escape as freely migrating defects; its value is that reported for proton irradiation. The vacancy and interstitial migration energies and pre-exponential factors are taken from the literature and adjusted within their reported ranges, jointly with the transfer

coefficients, to reproduce both the magnitude and the width of the measured grain-boundary composition profiles across the dose series. Because these profiles are measured before exposure, the RIS calibration is independent of the oxidation model. The resulting values are given in Table 1.

2.4.2 Oxidation Kinetics Parameterization

The oxidation model requires the interfacial rate constants k_{r}^{k} of the four oxide phases, their oxygen diffusivities D_{O}^{k}, the surface transfer coefficient k_{s} of the Robin condition Eq. (37), the magnetite threshold $C_{\mathrm{O}}^{\mathrm{th}}$, and the redistribution velocities v_i^{r} of Eq. (30). The oxide molar volumes and densities used in the thickness update are taken from crystallographic data and held fixed.

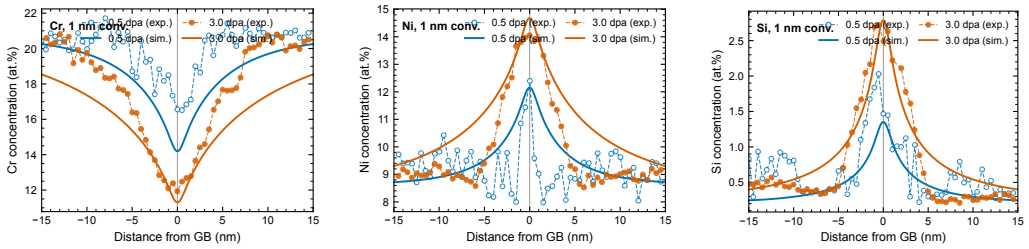
The model is calibrated against the dose-dependent grain-boundary oxidation depths of Deng et al. [5], measured after exposure in simulated PWR primary water at 0, 0.5, and 3 dpa. The predicted oxidation depth is defined as the distance from the boundary mouth to the deepest position at which the total oxide thickness $\sum_k L_k$ exceeds a fixed threshold L_{th}, applied identically at every dose. The free parameters are adjusted to reproduce the measured depths at all three doses simultaneously, using a single parameter set; no parameter is allowed to take different values at different doses. The resulting values are given in Table 2.

Table 1: Parameters of the radiation-induced segregation model.

Symbol	Description	Value	Source

Table 2: Parameters of the grain-boundary oxidation model.

Symbol	Description	Value	Source



(a) (b) (c)

Figure 3: Calculated and measured grain-boundary composition profiles after irradiation to 0.5 and 3.0 dpa: (a) Cr, (b) Ni, (c) Si. Solid lines are calculated profiles obtained with the finite-efficiency sink condition, Eqs. (13) and (14), and convolved with a 1 nm FWHM Gaussian; symbols are the measurements of Deng et al. [3] at 0.5 and 3.0 dpa.

3 Results

3.1 Radiation-Induced Segregation Profiles

Figure 3 compares the calculated and measured grain-boundary composition profiles after irradiation to 0.5 and 3.0 dpa. The calculated profiles are convolved with a 1 nm Gaussian to account for the finite spatial resolution of the measurements. These profiles are the target of the RIS calibration of Sec. 2.4.1.

Cr is depleted at the boundary and Ni and Si are enriched, at both doses. The calculated boundary Cr concentration falls from the matrix value of about 20 at.% to about 14 at.% at 0.5 dpa and about 11 at.% at 3.0 dpa, against measured minima of about 16.5 and 11.5 at.% (Fig. 3a). Ni rises from about 9 at.% to about 12 and 14.5 at.%, against measured maxima of about 12.5 and 14.5 at.% (Fig. 3b). Si rises from about 0.3 at.% to about 1.4 and 2.8 at.%, against measured maxima of about 2.0 and 2.8 at.% (Fig. 3c). The boundary values at 3.0 dpa are reproduced for all three elements; at 0.5 dpa the model overestimates the Cr depletion.

3.2 Oxide Composition and Interfacial Chemistry

Figure 4 compares the calculated composition across the intergranular oxide with the EDX line scans of Deng et al. [5] after the 500 h exposure. The

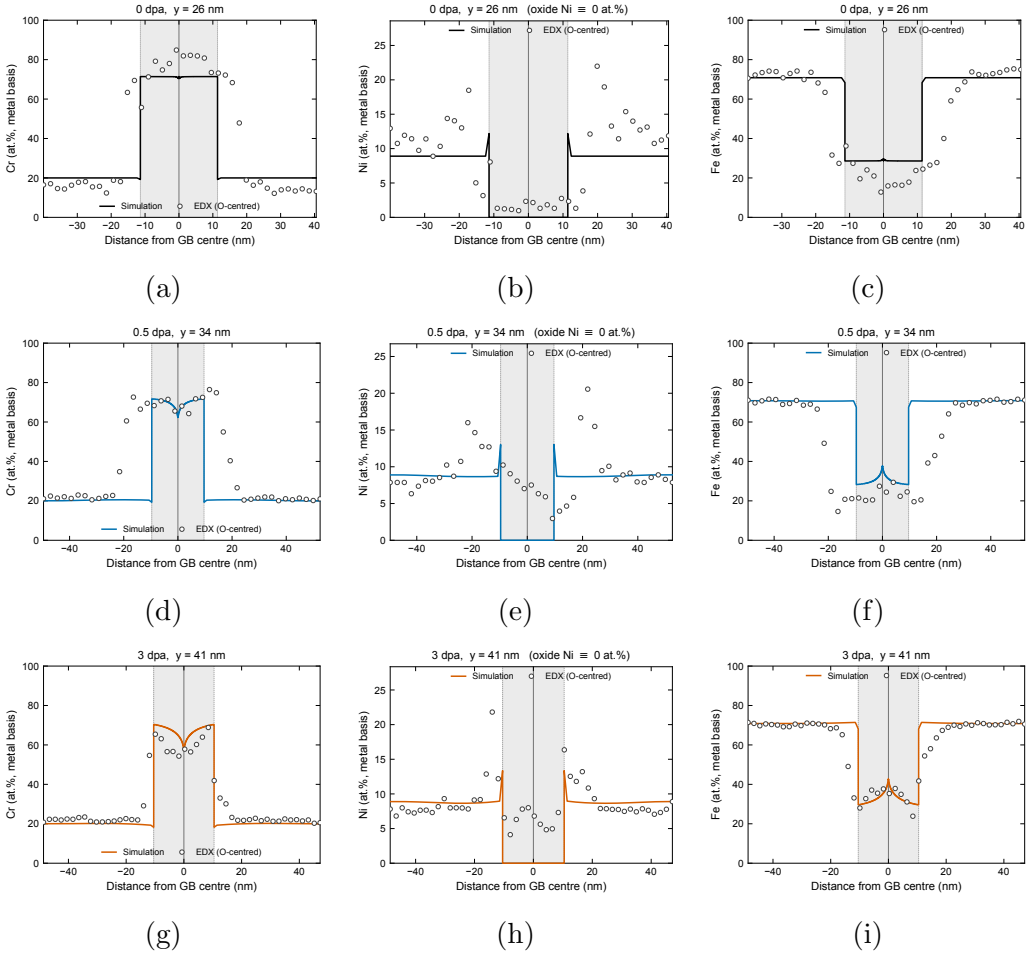


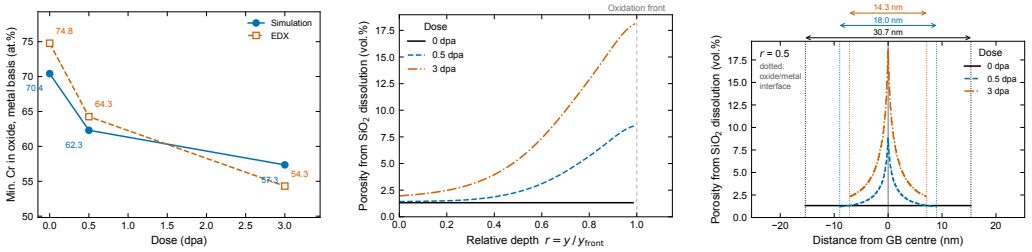
Figure 4: Calculated and measured composition profiles across the intergranular oxide after the 500 h exposure: (a–c) solution-annealed, (d–f) 0.5 dpa, (g–i) 3.0 dpa; Cr, Ni and Fe from left to right. Calculated profiles are convolved with a Gaussian of 1 nm FWHM to match the EDX probe; measurements are the line scans of Deng et al. [5].

calculated profiles are plotted with the oxide formed first at the centre of the intergranular oxide and the later oxide growing outward to the oxide/metal interface on both sides, consistent with symmetric growth into the two grains.

The measurements show the signature of selective oxidation at all three doses: Cr is enriched and Fe depleted in the oxide relative to the matrix. The Cr content of the oxide, normalized to exclude oxygen, decreases with

dose, from 74.9 at.% in the solution-annealed specimen to 69.5 and 58.3 at.% at 0.5 and 3.0 dpa [5]. On the metal side of the interface the scans show no significant Cr depletion or Fe enrichment, and only a slight Ni enrichment. In the irradiated specimens the Cr profile exhibits a minimum at the centre of the oxide that deepens with increasing dose.

The calculated profiles reproduce these features. With the interfacial mass redistribution of Eqs. (28)–(31), the calculated Cr profile retains the shape of the segregation profile across the interface, with neither Cr depletion nor Fe enrichment on the metal side, and shows the central Cr minimum whose depth increases with dose. The calculated Cr and Fe contents of the oxide follow the measured dose dependence: the oxide formed on irradiated boundaries is poorer in Cr and richer in Fe. The small amount of Ni in the oxide is carried by the Ni fraction λ_{Ni} of Eq. (30) as unoxidized solute; no Ni oxide forms in the model, consistent with Deng et al. [3]. The calculated SiO_2 volume fraction, which the model converts to free volume (Sec. 2.2), is reported with the oxide composition as the porosity generated within the intergranular oxide.



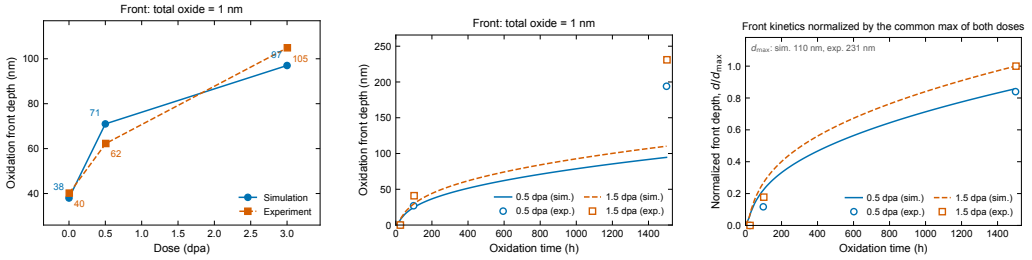
(a) (b) (c)

Figure 5: Comparing the Cr content and porosity created by SiO_2 dissolution volume fraction between SA sample and irradiated sample. (a) Comparing the calculated minimum of the Cr content in the intergranular oxide with that in experiments [5]. (b) Comparison of porosity volume fraction between different doses. (c) Comparison of the porosity volume fraction profile across the intergranular oxide section between different doses.

Figure 5a shows that the Cr content in the intergranular oxide decreases with doses, which is consistent with experimental results [5]. Assuming all formed SiO_2 is dissolved immediately, the intergranular oxide have higher porosity volume fraction in GB irradiated by higher doses which is showed

in figure 5b. Figure 5c shows that the porosity induced by silica dissolution is concentrated at the center of intergranular oxide, which is consistent with the observation of Kuang et al. [16] by under-focus TEM characterization.

3.3 Intergranular Oxide Penetration Depth



(a) (b) (c)

Figure 6: Intergranular oxide penetration depth, calculated and measured. (a) Dose dependence after 500 h; these data calibrate the oxidation parameters. Measurements from Deng et al. [5]. (b) Time dependence at 0.5 and 1.5 dpa, predicted with the calibrated parameters held fixed. Measurements from Deng et al. [4]. (c) As (b), with depths normalized by the maximum over both doses, $L/L_{\max}$, separately for calculations and measurements.

The oxidation parameters are calibrated against the dose dependence of the intergranular oxide penetration depth after 500 h of exposure measured by Deng et al. [5] at 0, 0.5 and 3.0 dpa, Fig. 6a. With a single parameter set the calculated depths are 38, 71 and 97 nm against measured values of 40, 62 and 105 nm: the unirradiated depth is matched, the 0.5 dpa depth is overestimated by 9 nm and the 3.0 dpa depth underestimated by 8 nm. The increase of penetration depth with dose is carried entirely by the segregation profiles inherited from the irradiation stage, Eq. (36).

3.4 Predictions against Independent Experiments

With the parameters fixed by the calibration of Fig. 6a, the model is used to predict the time dependence of the penetration depth, which is compared in Fig. 6 with the measurements of Deng et al. [4] at 0.5 and 1.5 dpa after 100 and 1500 h. At 100 h the calculated and measured depths are of the

same order, 20–40 nm. At 1500 h the model gives about 95 and 110 nm against measured values of about 194 and 231 nm, roughly half (Fig. 6b). The two experimental series are, however, not mutually consistent: although the authors report the same material, irradiation and exposure conditions, the apparent effective diffusivity of the oxidation front, $\bar{D}_{\mathrm{eff}} = L^2/2t$, is larger for the 0.5 dpa specimen of Deng et al. [4] than for the 3.0 dpa specimen of Deng et al. [5] (Supplementary Information, Table S1). A model calibrated on the 2017 series is therefore expected to underestimate the depths of the 2020 series. When the depths are normalized by the maximum over both doses, Fig. 6c, the calculated 0.5 dpa value at 1500 h, 0.85, matches the measured 0.84, so the ratio between the two doses at long times is reproduced; at 100 h the calculated normalized depths, about 0.20 and 0.25, exceed the measured 0.12 and 0.18, i.e. the calculated kinetics are relatively faster at short times. The model thus predicts the relative effect of dose at long exposure, while the absolute rate is tied to the series used for calibration.

3.5 Fully Coupled Predictions under Concurrent Irradiation and Oxidation

4 Discussion

4.1 Effect of Radiation-Induced Segregation on Grain-Boundary Oxidation

The composition profiles of Sec. 3.2 show that the Cr concentration at the advancing interface is set by the radiation-induced segregation profile and is not modulated by the selective oxidation: the shape of the segregation profile is retained as the front moves into the metal, and the central Cr minimum of the oxide is the segregation minimum of the boundary that was oxidized first. This retention is not reproduced by a model in which all unoxidized metal is rejected to the metal side. Because Cr reacts much faster than Fe, such a model depletes Cr and enriches Fe strongly ahead of the front, in conflict with the scans. The interfacial redistribution of Eqs. (28)–(31) resolves this by carrying the excess Fe and Ni swept by the front away through the oxide, along its grain boundaries to the surface, as proposed by Deng et al. [4] from the outer-layer oxide particles observed on irradiated specimens. The agreement with the measured profiles therefore supports the picture in which

selective oxidation consumes Cr at the rate the segregation profile supplies it, while the elements it does not consume leave through the oxide rather than accumulating in the metal.

4.2 Rate-Controlling Transport and Oxide Chemistry

Silica formed at the boundary is not retained. The segregated Si is oxidized preferentially and the resulting SiO_2 dissolves into the water, leaving a porous interface along the centre of the intergranular oxide [16]; the same dissolution degrades the inner oxide film on high-Si model alloys and accelerates intergranular oxidation [7]. The model adopts this behaviour as a limiting case: all SiO_2 produced is treated as dissolved and the volume it occupied is retained as free volume available for oxygen transport (Sec. 2.2). The calculated SiO_2 volume fraction reported in Sec. 3.2 is therefore a measure of the porosity generated within the intergranular oxide rather than of a phase present in it, and its contribution to $D_{\mathrm{O}}^{\mathrm{eff}}$ through Eq. (15) is the route by which Si enrichment enters the oxidation kinetics.

4.3 Implications of Concurrent Irradiation and Oxidation

4.4 Predictive Capability and Model Limitations

5 Conclusions

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